Shirpur Education Society's

R. C. PATEL INSTITUTE OF TECHNOLOGY, SHIRPUR

An Autonomous Institute

(Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere)

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Programme: B. Tech (AI & DS)/B. Tech (AI & ML)

Year: III/Semester V (Exam Year: 2025-2026)

Subject: Image Processing and Computer Vision **Time:** 09:30 am - 11:30 am (02:00 Hrs.)

Date: 05 Dec 2025 **Max Marks:** 60

END SEMESTER EXAMINATION ODD SEM-V-REGULAR

Instructions: 1. This question paper contains **3** pages

2. Answer to each new question to be started on a fresh page.

3. Figure in right hand side indicates full marks

1. All questions are compulsory **15**

a. Explain the Canny Edge Detection algorithm and describe its main stages. **5**

b. What are discontinuities in an image? Explain the three basic types: point, line, and edge discontinuities with examples. **5**

c. Explain the concept of neighborhood processing in image enhancement. Give one example of its application. **5**

2. **15**

a. Attempt any 1. **10**

i. Explain the Ideal Low-Pass Filter (ILPF) used in frequency domain image smoothing. Discuss its properties, advantages, and disadvantages. **10**

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ii. Given the 4×4 image (grayscale intensities) and the 4×4 Sylvester Hadamard matrix Compute the (unnormalized) 2-D Hadamard transform $Y=H_4 I H_4^T$. **10**

$$I = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix} \qquad H_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

b Consider following image: **5**

2	1	1	0	0	0
4	4	4	2	4	0
5	3	5	1	4	1
5	4	3	0	3	0
4	2	5	0	1	1
3	3	4	0	2	0

Explain how Thresholding can be performed using Otsu's Algorithm Perfrom only 1 Iteration.

3. **15**

a. Given an 8 x 8 image. Segment the image using the **Split and Merge** technique. The predicate to be used is that the difference in the intensity value of all the pixels in a region should not exceed 2. **7**

0	0	0	6	6	2	2	2
0	0	6	5	6	2	2	2
0	0	6	6	6	2	2	2
0	0	5	6	6	2	2	2
0	0	6	5	6	2	2	2
0	0	5	5	2	2	2	2
0	0	6	6	2	2	2	2
0	0	6	5	2	2	2	2

b. Attempt Any 1. **8**

i. Calculate IGS code for following Data; {200,201,250,251,252,253,254,255} **8**

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ii. Construct a Huffman tree and find the Huffman codes for the symbol set: **8**
{ P:0.3, Q:0.25, R:0.2, S:0.15, T:0.1 }

4. **15**

a. Attempt any 1. **7**

i. Perform the Histogram Equalization for following histogram of 8*8 image given below. Draw the histograms of the original and equalized images. Comment on the contrast of the image before because grain equalization and after histogram equalization. **7**

Gray Levels	0	1	2	3	4	5	6	7
No. of Pixels	9	8	11	4	10	15	4	3

---------- OR ----------

ii S is the given 8 bit image **7**

35	65	45	48	72
67	45	32	93	79
76	33	69	49	71
45	123	55	28	45
76	23	78	61	67

Perform:

1. Digital Negative
2. Thresholding at T = 30
3. Gray Level Slicing with and without background if r1 = 25, r2=85
4. Explain bit plane slicing and only give MSB and LSB plane of above image.

b. **8**

i. Given the input image: **8**

10	14	1	4	9
2	2	8	7	13
8	4	5	12	10
1	12	13	14	16
8	12	15	16	14

Apply Harris Corner Detector and find out if central 3 x 3 pixels are: Corner, Edge or Flat.

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ii. Why does the Histogram of Oriented Gradients technique use gradient orientations instead of raw pixel intensities for describing objects in an image? **8**